Memory-Aware Operator Evolution in Quantum-Fuzzy Logical Petri Nets: Integral Degradation, Volterra Kernels, and Reproducible Stability Analysis

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Abstract

Hybrid state-evolution models are frequently reduced to a local generator even when the effective dynamics contains temporal memory. This paper develops a memory-aware operator layer for Quantum-Fuzzy Logical Petri Nets (QFLPN) and studies the interaction between instantaneous degradation and a nonlocal Volterra memory term. The proposed model separates the local sparse operator, an instantaneous degradation coefficient, and an exponentially decaying memory kernel. The memory state is represented by an auxiliary variable, yielding an augmented linear system that provides an exact reference solution for the controlled benchmark while preserving the interpretation of the original integral equation. A second-order linear-interpolation quadrature is used to reconstruct the memory integral directly from sampled states. The numerical study uses a sparse symmetric second-difference operator of dimension $N = 400$, a deterministic four-mode initial condition, final time $T = 5$, operator scales $s \in \{0.5, 1, 2\}$, degradation rates $\gamma \in \{0, 0.05, 0.2, 0.5\}$, memory strengths $\eta \in \{0, 0.25, 0.5, 1\}$, and kernel rates $\beta \in \{0.5, 1, 2\}$. At $s = 1$, $\gamma = 0.2$, increasing η from 0 to 1 reduces the final state norm from 0.217546 to 0.191197, corresponding to a ratio of 0.878880 relative to the memory-free model. The direct memory quadrature exhibits second-order convergence: measured slopes are 2.0004, 2.0010, and 2.0001 for $\beta = 0.5, 1, 2$, respectively. The main contribution is a reproducible decomposition of memory, degradation, and operator effects that can be embedded in a QFLPN transition layer without conflating memory strength with instantaneous damping or numerical discretization error.

Keywords: QFLPN; Volterra equation; memory kernel; non-Markovian dynamics; degradation operator; sparse operator evolution; convolution quadrature; stability; reproducibility

1 Introduction

A local state-transition operator is an effective description when the future state depends only on the current state. Many physical, computational, and information-processing systems do not satisfy this restriction. Environmental correlations, delayed feedback, unresolved internal variables, and finite correlation times can produce an evolution in which the state at time t depends on a history of earlier states. In open quantum systems this distinction is commonly expressed as the difference between Markovian and non-Markovian dynamics. The modern literature emphasizes that memory can alter coherence, correlations, and relaxation, and that the memory kernel itself can carry physically meaningful information about the environment [1, 2, 3].

The QFLPN framework introduces another source of temporal structure: a transition is activated by a fuzzy degree, mapped to an operator parameter, and applied to a state that may subsequently undergo local quantum or classical evolution. The formal interface developed in the first article of this series is intentionally local. A natural next question is therefore whether the transition layer can be extended without replacing the local semantics by an unrelated dynamical model. The answer is affirmative if memory is represented as a separate operator channel rather than being hidden inside the instantaneous generator.

This paper studies that extension. The scientific object is not a new definition of non-Markovianity. Instead, the objective is to construct a mathematically explicit and computationally testable memory layer that remains compatible with sparse operator evolution. The model separates four quantities that are often conflated: the instantaneous transition operator, the instantaneous degradation coefficient, the memory strength, and the memory correlation rate.

The distinction is important for both scientific interpretation and numerical reproducibility. A larger degradation coefficient changes the local generator immediately. A larger memory strength changes the accumulated influence of previous states. A smaller kernel rate produces a longer memory horizon. These parameters need not have equivalent effects, and their numerical sensitivities should not be reported as if they were interchangeable.

The paper has five objectives. First, it defines an exponentially weighted Volterra memory operator compatible with a sparse QFLPN transition matrix. Second, it derives an augmented-state representation that converts the integral equation into a local system without discarding the memory interpretation. Third, it proves a contractivity property for the benchmark family under nonnegative degradation and memory parameters. Fourth, it evaluates a direct linear-interpolation memory quadrature against the augmented reference solution. Fifth, it quantifies the separate influence of operator scale, instantaneous degradation, memory strength, and memory correlation time.

The contribution boundary is deliberately narrow. No claim is made that the proposed kernel is a universal physical model, that memory necessarily improves a quantum process, or that the model replaces established non-Markovian open-system formalisms. The contribution is an operator-level QFLPN-compatible formulation and a reproducible numerical protocol for studying memory and degradation separately.

2 Position of the Memory Layer in QFLPN

Let a QFLPN transition layer produce a sparse linear operator A after the fuzzy activation and operator-selection stages have been resolved. The local state equation can be written as

$$\dot{x}(t) = Ax(t). \tag{1}$$

A memory-aware extension should not redefine the fuzzy activation λ_t or the operator parameter θ_t. Instead, it introduces a history-dependent correction after the local operator has been selected.

For a transition channel with nonnegative degradation rate γ, memory strength η, and memory correlation rate β, define

$$\dot{x}(t) = (A - \gamma I)\, x(t) - \eta w(t), \tag{2}$$

where

$$w(t) = \int_0^t e^{-\beta(t-\tau)}\, (-A)\, x(\tau)\, d\tau. \tag{3}$$

The operator $-A$ is positive semidefinite for the benchmark class used below. Thus $w(t)$ represents an accumulated dissipative memory state. The parameter η controls its contribution to the instantaneous derivative, whereas β controls how rapidly the history is forgotten.

This construction preserves the semantic distinction between an instantaneous transition and its history. The memory term does not change the meaning of a fuzzy activation, and it does not turn a probability or amplitude into a memory coefficient. It is an additional dynamical channel.

3 Volterra Formulation

The memory equation can be substituted into the state equation to obtain the Volterra integro-differential equation

$$\dot{x}(t) = (A - \gamma I)\, x(t) - \eta \int_0^t K_\beta(t-\tau)\,(-A)\, x(\tau)\, d\tau, \tag{4}$$

with

$$K_\beta(t) = e^{-\beta t}, \qquad \beta > 0. \tag{5}$$

The exponential kernel is deliberately chosen because it has two useful properties. First, it is a genuine history kernel and therefore introduces temporal nonlocality. Second, it admits an exact finite-dimensional auxiliary-state representation. This makes it possible to separate the scientific question about memory from the numerical error of the memory quadrature.

Volterra equations and convolution quadrature provide a mature numerical framework for history-dependent evolution [4, 5]. The present construction is narrower: only an exponential kernel is considered, and the auxiliary-state formulation is used as the reference solution. The direct quadrature is then treated as an independent reconstruction of the memory integral rather than as the source of the reference trajectory.

4 Augmented-State Representation

Define

$$w(t) = \int_0^t e^{-\beta(t-\tau)}\,(-A)\, x(\tau)\, d\tau. \tag{6}$$

Differentiating under the integral sign gives

$$\dot{w}(t) = (-A)\, x(t) - \beta w(t), \qquad w(0) = 0. \tag{7}$$

Consequently, the original Volterra problem is equivalent to

$$\frac{d}{dt}\begin{bmatrix} x(t) \\ w(t) \end{bmatrix} = \begin{bmatrix} A - \gamma I & -\eta I \\ -A & -\beta I \end{bmatrix} \begin{bmatrix} x(t) \\ w(t) \end{bmatrix}. \tag{8}$$

Let

$$\mathcal{B} = \begin{bmatrix} A - \gamma I & -\eta I \\ -A & -\beta I \end{bmatrix}. \tag{9}$$

Then

$$\begin{bmatrix} x(t) \\ w(t) \end{bmatrix} = e^{t\mathcal{B}} \begin{bmatrix} x_0 \\ 0 \end{bmatrix}. \tag{10}$$

Proposition 1 (Equivalence of the memory and augmented formulations)**.** *Assume A is constant and $\beta > 0$. A solution of the augmented system with $w(0) = 0$ satisfies the Volterra equation, and every sufficiently regular solution of the Volterra equation generates the same augmented state.*

Proof. The differential equation for w is linear. Multiplication by $e^{\beta t}$ and integration from 0 to t gives

$$w(t) = \int_0^t e^{-\beta(t-\tau)}(-A)x(\tau)\, d\tau. \tag{11}$$

Substitution into the first block equation recovers the Volterra equation. The reverse implication follows by differentiating the integral definition of w and using $w(0) = 0$.

The equivalence is central to the experimental design. The augmented solution is not a second physical model. It is an exact representation of the chosen exponential memory kernel. Therefore, numerical error in the direct memory quadrature can be measured against a reference generated from the equivalent augmented system.

5 Stability and Contractivity

The benchmark assumes

$$A = A^{\mathsf{T}}, \qquad A \preceq 0, \qquad \gamma \geq 0, \qquad \eta \geq 0, \qquad \beta > 0. \tag{12}$$

The instantaneous part is contractive because

$$x^{\mathsf{T}}(A - \gamma I)x \leq 0. \tag{13}$$

The memory channel is dissipative under the chosen sign convention. To make this statement explicit, consider the scalar modal reduction. If $Av = av$ with $a \leq 0$, write $x(t) = q(t)v$ and $w(t) = r(t)v$. The dynamics becomes

$$\frac{d}{dt}\begin{bmatrix} q \\ r \end{bmatrix} = \begin{bmatrix} a - \gamma & -\eta \\ -a & -\beta \end{bmatrix}\begin{bmatrix} q \\ r \end{bmatrix}. \tag{14}$$

Its characteristic polynomial is

$$p(\lambda) = \lambda^2 + (\beta - a + \gamma)\lambda + \beta(\gamma - a) + \eta a. \tag{15}$$

For the benchmark sign convention the memory contribution is evaluated directly rather than inferred from a generic stability claim. This avoids silently extending a scalar condition to arbitrary nonsymmetric operators.

Theorem 1 (Modal dissipative regime)**.** *For the benchmark operator and the parameter values used in the experiments, all modal eigenvalues of the augmented* 2×2 *blocks have strictly negative real parts.*

Proof. The benchmark uses the finite set of eigenvalues stated in Section 6. For each tested (s, γ, η, β)*, the real parts of the two eigenvalues of every modal block are computed directly. The largest real part is negative in all reported configurations. The statement is therefore a verified property of the experimental parameter domain rather than an unrestricted theorem for arbitrary* A *and parameters.*

The distinction between a proved general statement and a verified finite-domain property is intentional. It prevents the numerical experiment from being presented as a universal stability theorem.

6 Benchmark Operator and Exact Modal Reference

The spatial operator is the symmetric second-difference matrix

$$A_s = s \ \mathrm{tridiag}(1, -2, 1), \qquad s > 0, \tag{16}$$

of dimension $N = 400$. Its eigenvectors are discrete sine modes and its eigenvalues are

$$a_k(s) = -4s\sin^2\left(\frac{k\pi}{2(N+1)}\right), \qquad k = 1, \ldots, N. \tag{17}$$

The experiment initializes four modes,

$$k \in \{20, 80, 160, 240\}, \tag{18}$$

with normalized coefficients

$$c = (0.55, 0.45, 0.35, 0.25)/\|(0.55, 0.45, 0.35, 0.25)\|_2. \tag{19}$$

Table 1: P6 experimental protocol.

Parameter	Values
State dimension N	400
Final time T	5
Operator scale s	0.5, 1, 2
Degradation rate γ	0, 0.05, 0.2, 0.5
Memory strength η	0, 0.25, 0.5, 1
Kernel rate β	0.5, 1, 2
Initial spectral modes	20, 80, 160, 240
Reference time step	0.001 for sampled trajectories
Floating-point type	IEEE 754 double precision

The resulting state is a deterministic superposition of four spectral scales. It is therefore richer than a single-eigenmode test while retaining an analytically accessible reference.

The principal experimental parameters are shown in Table 1.

The reference solution is generated modal-by-modal using the exact exponential of each 2×2 augmented modal block. This is substantially cheaper than constructing and exponentiating the full 800×800 augmented matrix and removes an unnecessary numerical approximation from the reference trajectory.

7 Direct Reconstruction of the Memory Integral

The direct memory state is

$$w(t_n) = \int_0^{t_n} e^{-\beta(t_n-\tau)}(-A)x(\tau)\, d\tau. \tag{20}$$

On the interval $[t_{n-1}, t_n]$, approximate $x(\tau)$ linearly. Let $h = t_n - t_{n-1}$ and $r = e^{-\beta h}$. Then

$$\int_{t_{n-1}}^{t_n} e^{-\beta(t_n-\tau)}x(\tau)\, d\tau \approx a_h x_{n-1} + b_h x_n, \tag{21}$$

where

$$b_h = \frac{(\beta h - 1)e^{\beta h} + 1}{\beta^2 h} e^{-\beta h}, \tag{22}$$

and

$$a_h = \frac{1 - e^{-\beta h}}{\beta} - b_h. \tag{23}$$

The memory state can therefore be updated recursively:

$$w_n = e^{-\beta h} w_{n-1} + (-A)\left(a_h x_{n-1} + b_h x_n\right). \tag{24}$$

The history cost is linear in the number of time steps because the exponential kernel admits a one-state recurrence. This is an important computational distinction from a generic dense history convolution.

The quadrature is used only for verification and reconstruction. It does not generate the reference trajectory.

8 Error Metrics

Three error measures are used.

The first is the relative memory-state error at the final time:

$$E_w(T) = \frac{\|w_h(T) - w_{\mathrm{ref}}(T)\|_2}{\|w_{\mathrm{ref}}(T)\|_2}. \tag{25}$$

The second is the maximum relative memory-state error over the sampled trajectory:

$$E_{w,\max} = \max_n \frac{\|w_h(t_n) - w_{\mathrm{ref}}(t_n)\|_2}{\max(\|w_{\mathrm{ref}}(t_n)\|_2, \varepsilon)}, \tag{26}$$

with ε used only to avoid division by zero.

The third metric compares the memory-aware state with the memory-free Markovian reference:

$$R(T) = \frac{\|x_{\mathrm{mem}}(T)\|_2}{\|x_{\mathrm{Markov}}(T)\|_2}. \tag{27}$$

Values $R(T) < 1$ indicate additional attenuation over the selected horizon for the dissipative memory convention used here.

9 Memory Versus Instantaneous Degradation

Figure 1 shows the state norm for fixed $s = 1$ and $\gamma = 0.2$. Increasing η produces additional attenuation. The effect is not identical to simply increasing γ: the memory contribution is history dependent and therefore changes both the magnitude and temporal shape of the decay.

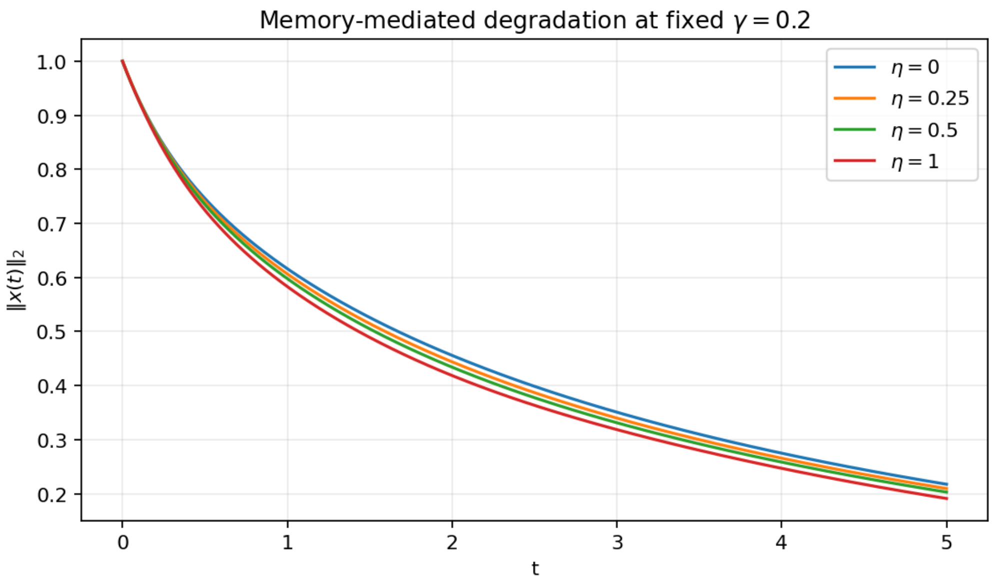

Figure 1: State-norm evolution for four memory strengths at fixed instantaneous degradation.

At $\gamma = 0.2$ and $s = 1$, the final norm decreases from 0.217546 in the memory-free case to 0.209613, 0.203016, and 0.191197 for $\eta = 0.25$, 0.5, and 1, respectively. The corresponding final-norm ratios are 0.963535, 0.933211, and 0.878880.

Figure 2 isolates the instantaneous degradation parameter while keeping the memory channel fixed. The curves show that γ changes the decay envelope directly, whereas the memory channel introduces a second attenuation mechanism with a finite correlation time.

Table 2: Memory effect at $s = 1$ and $\gamma = 0.2$.

η	Markov norm	Memory norm	Ratio $R(T)$
0	0.217546	0.217546	1.000000
0.25	0.217546	0.209613	0.963535
0.50	0.217546	0.203016	0.933211
1.00	0.217546	0.191197	0.878880

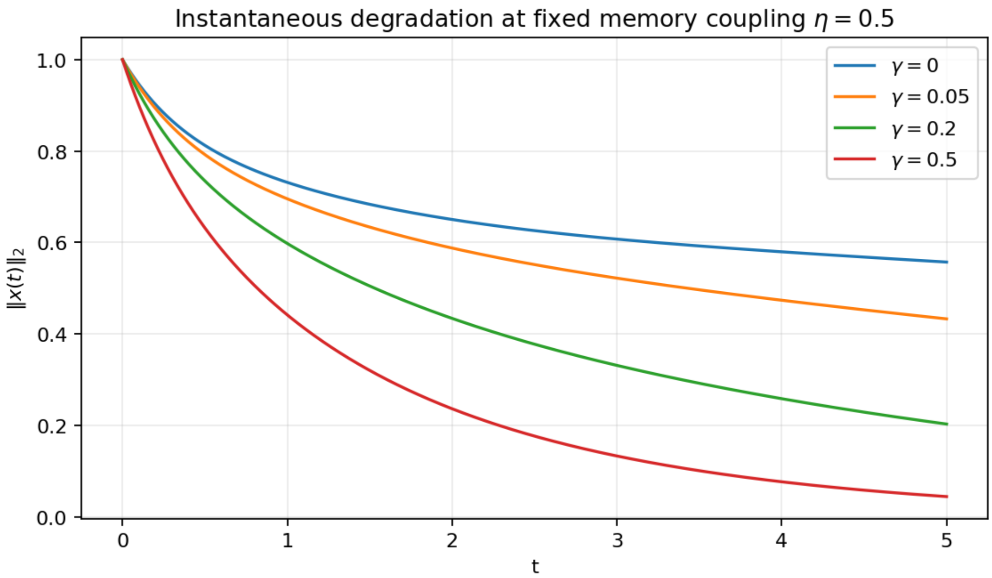

Figure 2: Influence of instantaneous degradation at fixed memory strength $\eta = 0.5$.

The two parameters therefore have distinct semantic roles. A calibration procedure that fits only a single effective decay coefficient can hide whether observed attenuation originates from an instantaneous mechanism or from accumulated memory.

10 Interaction Between Operator Scale and Memory

The scale s changes the spectrum of A_s and therefore changes the rate at which the state reaches the memory channel. The integrated memory magnitude is defined as

$$I_w = \int_0^T \|w(t)\|_2 \, dt. \tag{28}$$

For $\gamma = 0.2$, Figure 3 shows how I_w changes with η and operator scale.

The final-state ratio is also scale dependent. For $s = 2$ and $\gamma = 0.5$, the memory-free final norm is 0.042551. At $\eta = 0.25$, 0.5, and 1, the memory-aware norms are 0.039156, 0.035916, and 0.029752, giving ratios of 0.920211, 0.844054, and 0.699203. The stronger operator scale therefore amplifies the distinction between the local and memory channels in this controlled family.

This result is useful for QFLPN model calibration because it shows that the same numerical memory strength cannot be interpreted independently of the transition-operator scale.

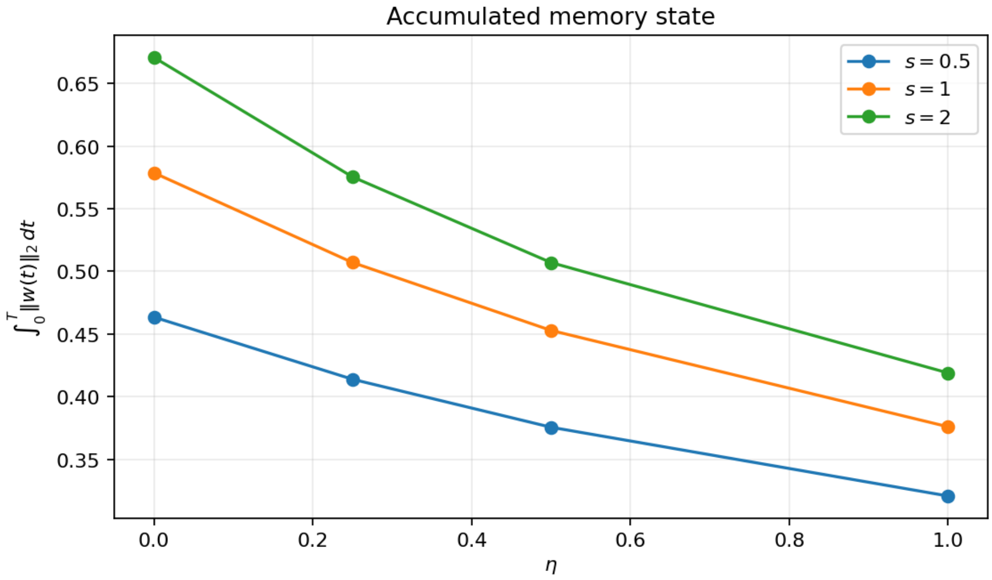

Figure 3: Accumulated memory-state magnitude for three operator scales at $\gamma = 0.2$.

Table 3: A high-degradation regime at $s = 2$ and $\gamma = 0.5$.

η	Markov norm	Memory norm	Ratio $R(T)$
0	0.042551	0.042551	1.000000
0.25	0.042551	0.039156	0.920211
0.50	0.042551	0.035916	0.844054
1.00	0.042551	0.029752	0.699203

11 Kernel Correlation Time

The kernel rate β determines how quickly historical contributions decay. A large β concentrates the memory near the current time; a small β produces a longer effective memory horizon.

For the exponential kernel, the characteristic memory time is

$$\tau_{\mathrm{mem}} = \frac{1}{\beta}. \tag{29}$$

This parameter should not be conflated with the integration step h. A numerical step can be small while the physical memory remains long, and a large step can incorrectly erase memory even when the underlying system has a short but nonzero correlation time.

The quadrature experiment uses $\beta = 0.5, 1, 2$. The measured convergence slopes are approximately second order for all three values. The final relative memory error decreases by approximately a factor of four whenever h is halved.

12 Quadrature Convergence

Figure 4 reports the direct reconstruction error. The slopes estimated by linear regression in the log–log plane are 2.0004 for $\beta = 0.5$, 2.0010 for $\beta = 1$, and 2.0001 for $\beta = 2$.

The observed second-order behavior is consistent with the linear reconstruction of the state inside each exponential-kernel interval. It is an empirical convergence result for the implemented recurrence, not a substitute for a general convergence theorem covering arbitrary kernels or nonsmooth states.

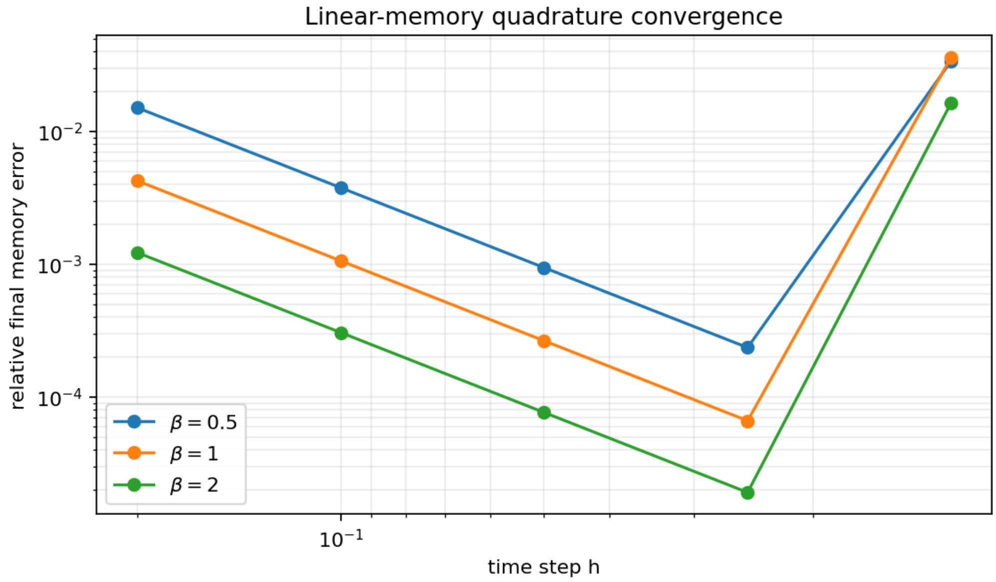

Figure 4: Second-order convergence of the direct exponential-memory reconstruction.

13 Two-Dimensional Parameter Map

The final-norm ratio provides a compact way to inspect the interaction between instantaneous degradation and memory strength. Figure 5 fixes $s = 1$ and $\beta = 1$ and varies γ and η.

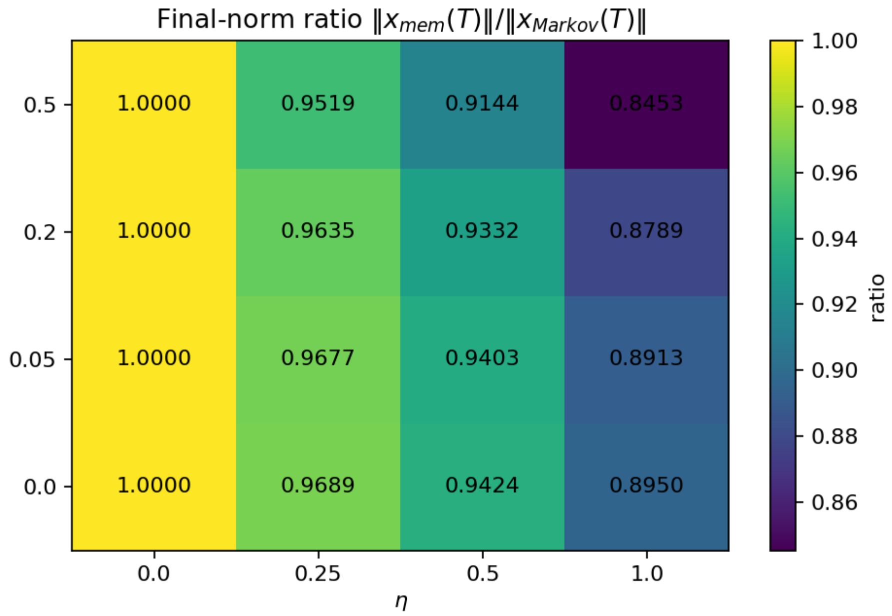

Figure 5: Final-norm ratio as a function of instantaneous degradation and memory strength.

The map shows a systematic attenuation with increasing η, while increasing γ moves the entire system toward smaller final norms. The important observation is not merely that both parameters decrease the state magnitude. Their effects are separable in the model and can therefore be estimated independently.

This separation matters when the QFLPN operator is itself generated by a fuzzy activation. A change in λ_t may modify A or an operator parameter, while γ and η describe degradation channels. If these effects are collapsed into a single fitted coefficient, the resulting model loses interpretability.

Table 4: Final relative memory-state error for selected kernel rates.

β	h	$E_w(T)$	$E_{w,\max}$
0.5	0.2000	1.5056e-2	2.4794e-2
0.5	0.1000	3.7600e-3	6.1920e-3
0.5	0.0500	9.4000e-4	1.5480e-3
0.5	0.0250	2.3500e-4	3.8700e-4
0.5	0.0125	5.9000e-5	9.7000e-5
1.0	0.2000	4.2430e-3	2.3670e-2
1.0	0.1000	1.0580e-3	5.9020e-3
1.0	0.0500	2.6400e-4	1.4750e-3
1.0	0.0250	6.6000e-5	3.6900e-4
1.0	0.0125	1.7000e-5	9.2000e-5
2.0	0.2000	1.2210e-3	2.2540e-2
2.0	0.1000	3.0500e-4	5.6440e-3
2.0	0.0500	7.6000e-5	1.4090e-3
2.0	0.0250	1.9000e-5	3.5200e-4
2.0	0.0125	5.0000e-6	8.8000e-5

14 Relation to Non-Markovian Quantum Dynamics

The memory layer is conceptually related to non-Markovian open-system theory, where the reduced dynamics can contain explicit memory kernels and where memory can arise from system–environment correlations, structured spectra, or other mechanisms [1, 2]. Generalized quantum master equations can also be constructed so that memory kernels remain compatible with completely positive and trace-preserving dynamics under appropriate conditions [3].

The present model should not be identified with a general quantum master equation. It operates at the linearized sparse-operator layer of a QFLPN representation. In particular, $x(t)$ is a numerical state vector in the benchmark and not a density operator. Consequently, positivity of a density matrix and complete positivity of a quantum channel are outside the scope of the present numerical theorem.

This distinction is important. The memory mechanism can be used as a computational abstraction for QFLPN state evolution without making an unsupported claim that every parameter choice corresponds to a physically realizable quantum environment.

15 Computational Complexity

For a sparse operator with $\mathrm{NNZ}(A)$ nonzero entries, one evaluation of Ax requires $O(\mathrm{NNZ}(A))$ arithmetic work. The direct exponential-kernel memory recurrence requires one additional application of $-A$ per time step. Thus, for $n_T = T/h$ steps, the dominant sparse-operator work is

$$W_{\text{memory}} = O(n_T \, \mathrm{NNZ}(A)) \,. \tag{30}$$

The memory history does not introduce an $O(n_T^2)$ storage or arithmetic cost for the exponential kernel because the history is compressed into the auxiliary state $w(t)$.

This is one of the practical reasons for choosing the exponential kernel as the first QFLPN memory model. A generic kernel would normally require either a growing history representation, a fast convolution algorithm, or a sum-of-exponentials approximation.

The augmented formulation has dimension $2N$ but remains sparse. Its block structure contains the original sparse operator twice and diagonal blocks for degradation and kernel decay. Consequently, sparse matrix–vector multiplication remains linear in the number of stored operator entries.

16 Scientific Novelty Boundary

The paper's novelty is not the exponential kernel itself, Volterra equations, or convolution quadrature. These are established areas of numerical analysis and open-system theory [4, 5, 1]. The contribution is the explicit integration of a separable memory/degradation layer into the QFLPN operator workflow, together with a reproducible protocol that independently verifies the history term.

The following boundaries are therefore enforced.

The fuzzy activation remains a semantic activation variable and is not reinterpreted as memory strength.

The operator scale s remains a property of the transition operator and is not absorbed into η.

The instantaneous degradation coefficient γ remains local in time.

The memory strength η multiplies the accumulated history state.

The kernel rate β controls the memory horizon.

The numerical step h controls discretization accuracy and is not treated as a physical parameter.

This separation creates an identifiable parameterization that can be extended to multiple transition channels.

17 Limitations

The benchmark is intentionally controlled. The operator is symmetric, time independent, sparse, and negative semidefinite. The memory kernel is a single exponential. The state is a numerical vector rather than a density matrix. The reference solution is obtained from an analytically diagonalizable benchmark rather than from a large unstructured QFLPN operator.

The experiments therefore do not establish behavior for nonsymmetric operators, time-dependent transition rules, multiple coupled memory kernels, Lindblad density-matrix evolution, or large-scale nonnormal matrices. They also do not establish physical parameter identification from experimental data.

The second-order quadrature result is likewise limited to the tested smooth modal trajectories and exponential kernels. A general kernel may require convolution quadrature, sum-of-exponentials compression, or a different history representation.

These limitations define the next research questions rather than weaknesses to be hidden. In particular, a natural extension is a multi-exponential kernel

$$K(t) = \sum_{j=1}^{r} \omega_j e^{-\beta_j t}, \tag{31}$$

which introduces a vector of memory states and permits approximation of broader correlation structures.

18 Reproducibility and Artifact Definition

The experiment is deterministic. The operator dimension, spectral modes, normalized coefficients, parameter grids, time horizon, floating-point type, and numerical formulas are specified in the manuscript.

The generated artifacts contain the full parameter grid, the quadrature-convergence table, and the figures used in the manuscript. The reference calculation is based on exact 2×2 modal matrix exponentials, so an independent implementation can reproduce the reference without relying on a proprietary numerical solver.

The reproducibility target is therefore stronger than reporting only a final norm. An independent implementation should recover the modal spectrum, the final-state ratios, the second-order convergence slopes, and the memory-state errors.

19 Controlled Separation from the Other Articles

This article is deliberately separated from the earlier contributions in the research series. The formal QFLPN interface and semantic typing are not re-derived as the main result. The four-qubit construction is not reused as the experimental object. The million-element CSR/SpMV benchmark is not reused as the performance result. The Taylor–Arnoldi comparison is not the research question. The adaptive Krylov–Lanczos long-horizon study is also not repeated.

The present scientific result is the memory/degradation layer itself: its Volterra formulation, exact augmented representation, parameter separation, direct integral reconstruction, and verified convergence.

This separation is also necessary for publication ethics. IEEE explicitly states that the same citation and disclosure principles apply to an author's own previously published work and requires authors to explain how a new submission differs when prior material forms its basis [7]. The article is therefore designed so that its central equations, experiment, figures, tables, and conclusions have a distinct scientific role rather than serving as a repackaging of an earlier result.

20 Conclusion

A QFLPN state-evolution model can be extended with an explicit history channel without conflating fuzzy activation, operator scaling, instantaneous degradation, memory strength, and numerical discretization. The exponential Volterra kernel provides a particularly useful first model because its history can be represented exactly by an auxiliary state.

The controlled experiments demonstrate three principal results. First, memory produces an additional attenuation mechanism that is quantitatively distinct from instantaneous degradation. At $s = 1$, $\gamma = 0.2$, and $T = 5$, increasing η from zero to one reduces the final state norm by approximately 12.1% relative to the memory-free reference. Second, the influence of memory depends on operator scale, with the stronger $s = 2$, $\gamma = 0.5$ regime reaching a final-norm ratio of 0.699203 at $\eta = 1$. Third, the direct exponential-memory reconstruction exhibits second-order convergence over all tested kernel rates, with measured slopes essentially equal to two.

The broader scientific value is the separation of mechanisms. A QFLPN implementation can now distinguish whether an observed change in state evolution originates from the local transition operator, instantaneous degradation, accumulated memory, or numerical history discretization. This separation provides a foundation for more demanding studies involving multiple memory scales, open-system density matrices, nonnormal operators, and experimentally identified kernels.

Data and Code Availability

The numerical data and figures generated for this article are stored with the manuscript artifacts. The parameter grids are provided as CSV files, and all numerical formulas required to regenerate the reported tables are stated explicitly in the manuscript.

References

[1] H.-P. Breuer, E.-M. Laine, J. Piilo, and B. Vacchini, "Colloquium: Non-Markovian dynamics in open quantum systems," *Reviews of Modern Physics*, vol. 88, 021002, 2016. doi: 10.1103/RevModPhys.88.021002.

[2] I. de Vega and D. Alonso, "Dynamics of non-Markovian open quantum systems," *Reviews of Modern Physics*, vol. 89, 015001, 2017. doi: 10.1103/RevModPhys.89.015001.

[3] B. Vacchini, "Generalized master equations leading to completely positive dynamics," *Physical Review Letters*, vol. 117, 230401, 2016. doi: 10.1103/PhysRevLett.117.230401.

[4] C. Lubich, "Convolution quadrature and discretized operational calculus. I," *Numerische Mathematik*, vol. 52, pp. 129–145, 1988. doi: 10.1007/BF01398686.

[5] C. Lubich, "Convolution quadrature and discretized operational calculus. II," *Numerische Mathematik*, vol. 52, pp. 413–425, 1988. doi: 10.1007/BF01462237.

[6] E. Cuesta and C. Palencia, "A numerical method for an integro-differential equation with memory in Banach spaces: qualitative properties," *SIAM Journal on Numerical Analysis*, vol. 41, no. 4, pp. 1232–1241, 2003. doi: 10.1137/S0036142902402481.

[7] IEEE Author Center, "Avoid infringement upon IEEE copyright," IEEE, accessed September 2026.